

The half-window of the powers of 3: a proof of Karttunen's conjecture on OEIS A037097, in the corrected form

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Abstract

OEIS A037097 reads one period of the n -th binary column of the powers of 3 — a window of length 2^{n-1} — as a polynomial over $\text{GF}(2)$. A. Karttunen conjectured in January 1999 that this polynomial is divisible by a large power of $x + 1$. The exponent as originally posted was $2^{n-1} - 1$; it was corrected to $2^{n-2} - 1$ on the entry by Q. Gu on 24 July 2026. We prove the corrected statement, and show the exponent is exact:

$$a(n) = (x + 1)^{2^{n-2}-1} (B(x)(x + 1) + x^{2^{n-2}}), \quad n \geq 3,$$

where B is the first quarter of the doubled window. We also confirm that the pre-correction exponent was indeed too large, already at $n = 3$. The proof rests on one 2-adic fact: $3^{2^{n-2}} \equiv 1 + 2^n \pmod{2^{n+1}}$, so advancing the exponent by 2^{n-2} complements bit n .

1 The sequence and the conjecture

OEIS A037097 (Antti Karttunen, 29 January 1999), *Periodic vertical binary vectors of powers of 3, starting from bit-column 2 (halved)*, has offset 2 and

$$a(n) = \sum_{k=0}^{2^{n-1}-1} \left(\lfloor 3^k / 2^n \rfloor \bmod 2 \right) 2^k, \quad a(2), \dots, a(5) = 0, 12, 120, 57120.$$

Identifying an integer with the $\text{GF}(2)$ polynomial whose coefficient of x^k is bit k , the entry carries

Conjecture: For $n \geq 3$, each term $a(n)$, when considered as a $\text{GF}(2)[X]$ -polynomial, is divisible by the $\text{GF}(2)[X]$ -polynomial $(x + 1)^{A000225(n-2)}$ ($= A051179(n - 2)$). [corrected by Qiuyan Gu, Jul 24 2026]

Here $A000225(n - 2) = 2^{n-2} - 1$; the parenthetical is consistent, since $(x + 1)^{2^m-1} = 1 + x + \dots + x^{2^m-1}$ encodes as the integer $2^{2^m} - 1 = A051179(m)$. As of the “Last modified” line on the live entry (23 August 2026) the statement is still recorded as an unproven conjecture, with no proof in the entry's links or programs.

Throughout $n \geq 3$ is fixed,

$$q = 2^{n-2}, \quad b(k) = \lfloor 3^k / 2^n \rfloor \bmod 2, \quad a(n) \longleftrightarrow A(x) = \sum_{k=0}^{2q-1} b(k)x^k,$$

so the window has length $2^{n-1} = 2q$. For nonzero $P \in \text{GF}(2)[x]$ let $v(P)$ denote the exponent of $x+1$ in P . We use $x^{2^m} + 1 = (x+1)^{2^m}$, the Frobenius endomorphism in characteristic 2.

2 The flip

Lemma 1. *For $m \geq 1$, $v_2(3^{2^m} - 1) = m + 2$; hence $3^{2^m} \equiv 1 + 2^{m+2} \pmod{2^{m+3}}$, and the multiplicative order of 3 modulo 2^{m+3} is 2^{m+1} .*

Proof. Lifting the exponent at $p = 2$ with even exponent N gives $v_2(3^N - 1) = v_2(3 - 1) + v_2(3 + 1) + v_2(N) - 1 = 1 + 2 + v_2(N) - 1$; take $N = 2^m$. Writing $3^{2^m} - 1 = 2^{m+2}u$ with u odd gives the congruence, and the order statement follows since $3^{2^m} \not\equiv 1$ but $3^{2^{m+1}} \equiv 1$ modulo 2^{m+3} . $\square$

With $m = n - 2 \geq 1$, read modulo 2^{n-1} : the order of 3 is $2^{n-1} = 2q$, so b is well defined on $\mathbb{Z}/2q\mathbb{Z}$ and the window is exactly one period.

Lemma 2 (the flip). *$b(k + q) = 1 - b(k)$ for every k .*

Proof. Bit n is determined modulo 2^{n+1} . By Lemma 1, $3^{k+q} \equiv 3^k(1 + 2^n) = 3^k + 2^n 3^k \equiv 3^k + 2^n \pmod{2^{n+1}}$, the last step because 3^k is odd. Adding 2^n modulo 2^{n+1} complements bit n and leaves the lower bits unchanged. $\square$

3 The theorem

Theorem 3. *Let $n \geq 3$, $q = 2^{n-2}$, and let $B(x) = \sum_{k=0}^{q-1} b(k)x^k$ be the first half of the window. Then*

$$A(x) = (x+1)^{q-1} \left(B(x)(x+1) + x^q \right),$$

and the second factor is not divisible by $x+1$. Hence $v(a(n)) = q - 1 = 2^{n-2} - 1 = A000225(n-2)$ exactly, and in particular $(x+1)^{A000225(n-2)}$ divides $a(n)$.

Proof. Put $S(x) = 1 + x + \cdots + x^{q-1}$. By Lemma 2 the window of length $2q$ consists of the block B followed by its complement, and complementing a 0/1 block of length q is adding S over $\text{GF}(2)$. Hence

$$A = B + x^q(B + S) = B(1 + x^q) + Sx^q.$$

Since q is a power of 2 we have $1 + x^q = (1 + x)^q$, and therefore $S = (1 + x^q)/(1 + x) = (1 + x)^{q-1}$. Substituting,

$$A = B(1 + x)^q + (1 + x)^{q-1}x^q = (1 + x)^{q-1} \left(B(1 + x) + x^q \right).$$

Evaluating the cofactor at $x = 1$ over $\text{GF}(2)$ gives $B(1) \cdot (1 + 1) + 1^q = 0 + 1 = 1 \neq 0$, so $x+1$ does not divide it, and $v(A) = q - 1$. $\square$

Proposition 4 (the pre-correction exponent was too large). *The statement obtained by replacing A000225($n - 2$) with A000225($n - 1$) = $2^{n-1} - 1$ is false for every $n \geq 3$. The smallest instance is $n = 3$: there $a(3) = 12$, whose GF(2) polynomial is $x^3 + x^2 = x^2(x+1)$, so $v(a(3)) = 1$, whereas $2^{n-1} - 1 = 3$.*

Proof. By Theorem 3, $v(a(n)) = 2^{n-2} - 1 < 2^{n-1} - 1$ for every $n \geq 3$. □

Remark (relation to the companion entry). A037096 takes the same column over a window of length 2^n , i.e. two periods, and its own conjecture asks for the exponent $3 \cdot 2^{n-2} - 1$; that is proved in a companion note by the same flip lemma. The two are consistent: the doubled window is $B, \overline{B}, B, \overline{B}$, giving $A_{A037096} = (x+1)^{3q-1}(B(x)(x+1) + x^q)$, and dividing by $(x+1)^{2q} = x^{2^{n-1}} + 1$ recovers Theorem 3. This matches the crossref $a(n) = \lfloor A037096(n)/2^{2^{n-1}} \rfloor$ recorded on the entry, and the identity $A037096(n) = (2^{2^{n-1}} + 1)a(n)$, both of which the verification script confirms.

Remark (honest assessment). Five lines once the flip lemma is in place, and the flip lemma is lifting the exponent. The natural destination is an OEIS comment. Note the peculiar status of this target: the conjecture in its currently posted form is one month old, having been corrected in July 2026, though the underlying claim dates to 1999. Proposition 4 confirms that correction was necessary and gives the smallest witness.

4 Verification

A verification script, written separately from this note, checks every claim and prints every number quoted. It (i) recomputes $a(2), \dots, a(9)$ from the defining sum and matches the published DATA exactly, including $a(3) = 12$, $a(4) = 120$, $a(5) = 57120$; (ii) verifies the factorization of Theorem 3 by explicit polynomial multiplication for $3 \leq n \leq 15$; (iii) computes $v(a(n))$ for $3 \leq n \leq 15$, obtaining 1, 3, 7, 15, 31, 63, 127, 255, 511, 1023, 2047, 4095, 8191, which is $2^{n-2} - 1$ in every case, with cofactor value 1 at $x = 1$ throughout, confirming exactness; and (iv) confirms the two crossref identities $a(n) = \lfloor A037096(n)/2^{2^{n-1}} \rfloor$ and $A037096(n) = (2^{2^{n-1}} + 1)a(n)$ for $3 \leq n \leq 12$.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A037097>, entry A037097, consulted 24 August 2026; also A037096, A000225 and A051179.
- [2] P. Ribenboim, *The New Book of Prime Number Records*, 3rd ed., Springer, 1996 (for the lifting-the-exponent lemma at $p = 2$).

A Suggested OEIS comment (plain text, house style)

The conjecture (in the corrected form, exponent A000225($n-2$)) is true, and the exponent is exact. Fix $n \geq 3$ and put $q = 2^{n-2}$, so the window has length $2^{n-1} = 2q$. By lifting the exponent, $v_2(3^{2^m} - 1) = m + 2$, so with $m = n-2$ we get $3^q = 1 + 2^n \pmod{2^{n+1}}$. Since 3^k is odd, $3^{k+q} = 3^k + 2^n 3^k = 3^k + 2^n \pmod{2^{n+1}}$, and adding 2^n modulo

$2^{(n+1)}$ complements bit n . So $b(k+q) = 1 - b(k)$: the window is a block B of length q followed by its complement. Over $GF(2)$, complementing a length- q block is adding $S = 1 + x + \dots + x^{(q-1)} = (x+1)^{(q-1)}$, so

$$a(n) = B(1 + x^q) + S*x^q = (x+1)^{(q-1)} * (B(x)(x+1) + x^q),$$

using $1 + x^q = (x+1)^q$ for q a power of 2. The cofactor at $x = 1$ is $B(1)*0 + 1 = 1$, so $x+1$ does not divide it: $v_{(x+1)}(a(n))$ is exactly $2^{(n-2)} - 1$. This also confirms that the pre-July-2026 exponent $A000225(n-1) = 2^{(n-1)} - 1$ was too large; the smallest witness is $n = 3$, where $a(3) = 12 = x^2(x+1)$ has only one factor of $x+1$.